

# AYMANE AIT DADS

Multimodal Sensing Intern | PyTorch · Audio ML · Computer Vision

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## Hiring Team - Intern, Multimodal Sensing for Smart Home Applications

Robert Bosch (South East Asia) Pte Ltd · Singapore

Dear Bosch Multimodal Sensing Hiring Team,

Multimodal sensing for smart home applications demands models that operate reliably across signal types - exactly the challenge I tackled building a **transformer autoencoder for unsupervised anomalous sound detection** on industrial audio, achieving **AUC 0.80** with no labeled anomalies, while separately ranking **#1 on the EURECOM leaderboard** for robust AI-generated image detection by fine-tuning a 428M-parameter vision-language model across 25+ generator types.

Fusing sensing modalities requires both signal processing depth and vision modeling rigor. For audio, I engineered **Mel spectrogram** pipelines with **SpecAugment** augmentation and benchmarked Mel vs. MFCC representations against a DCASE-style evaluation protocol. For vision, I designed **10-view test-time augmentation** on **CLIP ViT-L/14** fine-tuned on **250K samples**, running ablation studies across transformer unfreezing depth and learning rate schedules to prove generalization over overfitting - the kind of cross-modal, experiment-driven discipline a multimodal sensing role demands.

Bosch's Corporate Research setup in Singapore sits at the intersection of sensing hardware and applied AI for real home environments - a product context, not a benchmark context. I want to contribute to that pipeline directly. I would welcome the chance to discuss how my work translates to your team's sensing challenges.

Sincerely,

AYMANE AIT DADS